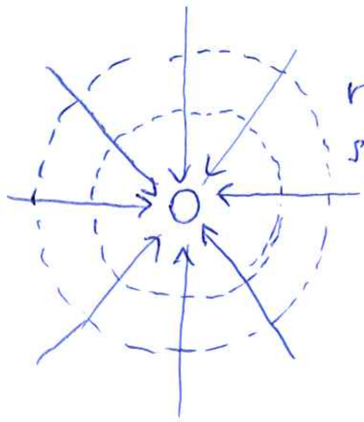
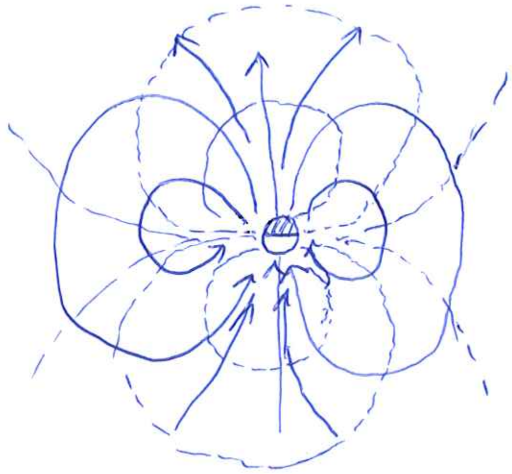
MAGNETICS

GRAVITY IS A MONOPOLAR FIELD



MAGNETICS IS A DIPOLAR FIELD

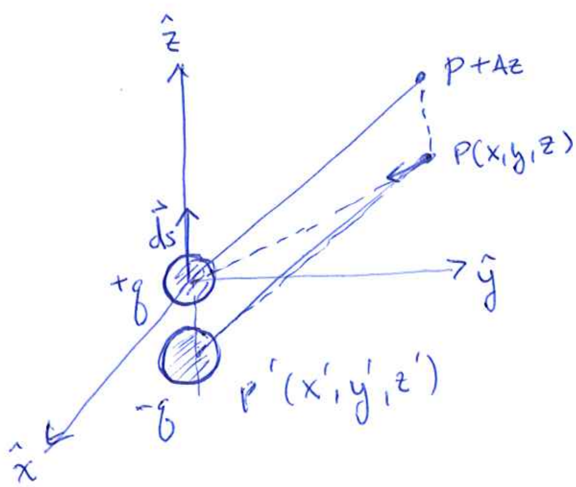


FIELD IS AXIALLY SYMMETRIC

DERIVING DIPOLAR FIELD

CONSIDER A MAGNETIC MONO POLE

LIKE A MAGNETIC MASS



BY SUPERPOSITION,

$$V(P) = V_1(P) + V_2(P)$$

$$= -\Big[\underset{+q}{V_1(P+\Delta z)} - \underset{-q}{V_2(P)}\Big]$$

IF Δz IS SMALL, WE CAN TURN ABOVE INTO A DERIVATIVE

$$V(P) = -\Delta z \frac{d V_1(P)}{dz}$$

BY ANALOGY TO GRAVITY

$$U(P) = G \frac{m}{r}$$ gravitational potential

$$V(P) = \underbrace{\frac{\mu_0}{4\pi}}_{\text{"fudge factor"}} \frac{q}{r}$$ magnetic potential (of monopole)

μ_0 - magnetic ~~potential~~ permeability of free space.

insert monopole potential into above

$$V(P) = -\frac{\mu_0}{4\pi} q \Delta z \frac{d}{dz} \frac{1}{r}$$

ORIENT AXES WITH DIPOLE AXIS AND MOVE CLOSE TOGETHER

$$V(P) = -\frac{\mu_0}{4\pi} q \, d\vec{s} \cdot \nabla_p \frac{1}{r}$$

gradient in potential at point P

DEFINE THE DIPOLE moment, $\vec{m} = q \, d\vec{s}$

$$V(P) = -\frac{\mu_0}{4\pi} \vec{m} \cdot \nabla_P \frac{1}{r}$$, magnetic potential of single dipole.

MAGNETIC FIELD IS RELATED TO POTENTIAL

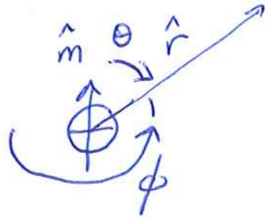
$$\vec{B} = -\nabla V$$

gradient

potential

magnetic induction / flux density

$$\vec{B}(r,\theta) = \frac{\mu_0}{4\pi}\frac{m}{r^3}\left[3(\hat{m}\cdot\hat{r})\hat{r} + \hat{m}\right]$$



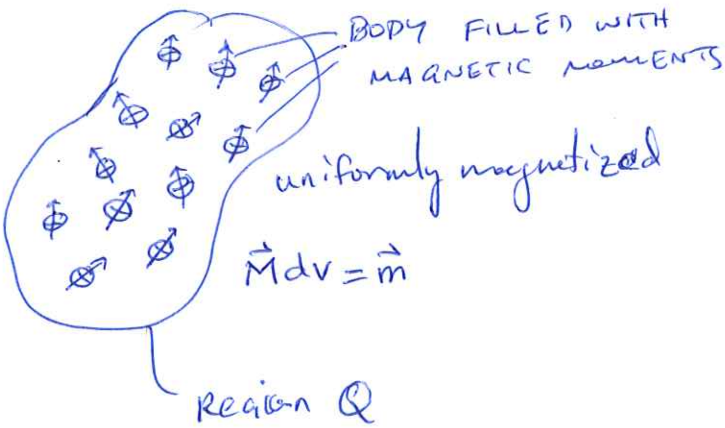
$$\hat{m}\cdot\hat{r} = \cos\theta$$

$$\vec{B}(r,\theta) = \frac{\mu_0}{4\pi}\frac{m}{r^3}\left[3\cos\theta\,\hat{r} + \hat{m}\right]$$

Can separate into a radial and azimuthal field

$$\boxed{\begin{aligned} B_r &= -\frac{\partial V}{\partial r} = \frac{\mu_0}{2\pi}\frac{m\cos\theta}{r^3} && \text{radial} \\ B_\theta &= -\frac{1}{r}\frac{\partial V}{\partial \theta} = \frac{\mu_0}{4\pi}\frac{m\sin\theta}{r^3} && \text{azimuthal} \\ |\vec{B}| &= \frac{\mu_0}{4\pi}\frac{m}{r^3}\left[3\cos^2\theta + 1\right]^{1/2} && \text{magnitude} \end{aligned}}$$

CAN USE THESE TO DEVELOP FIELD OF A BODY



$$\vec{M}\,dv = \vec{m}$$

$$\underbrace{\vec{M} = \frac{1}{V}\sum_i^N m_i}_{\text{integrate over volume}}$$

$$\vec{V}(P) = \frac{\mu_0}{4\pi} \nabla_P \int_V \vec{M}(Q) \cdot \nabla_Q \frac{1}{r}\, dv$$

$$\vec{B}(P) = -\frac{\mu_0}{4\pi} \nabla_P \int_R M \cdot \nabla_Q \frac{1}{r}\, dv$$

BUT HOW DOES MAGNETIZATION RELATE TO PHYSICAL PROPERTIES OF ROCKS?

$$\vec{H} = \frac{1}{\mu_0} \vec{B} - \vec{M}$$

magnetic field strength

FIELD STRENGTH $\vec{H}$ IS PROPORTIONAL TO MAGNETIZATION $\vec{M}$

$$\vec{M} = \chi \vec{H}$$

χ - SUSCEPTIBILITY (ABILITY TO BE MAGNETIZED)

$$\vec{H} = \frac{1}{\mu_0} \vec{B} - \chi \vec{H}$$

$$\vec{H}(1+\chi) = \frac{1}{\mu_0} \vec{B}$$

$$\vec{B} = \mu_0 \vec{H}(1+\chi)$$

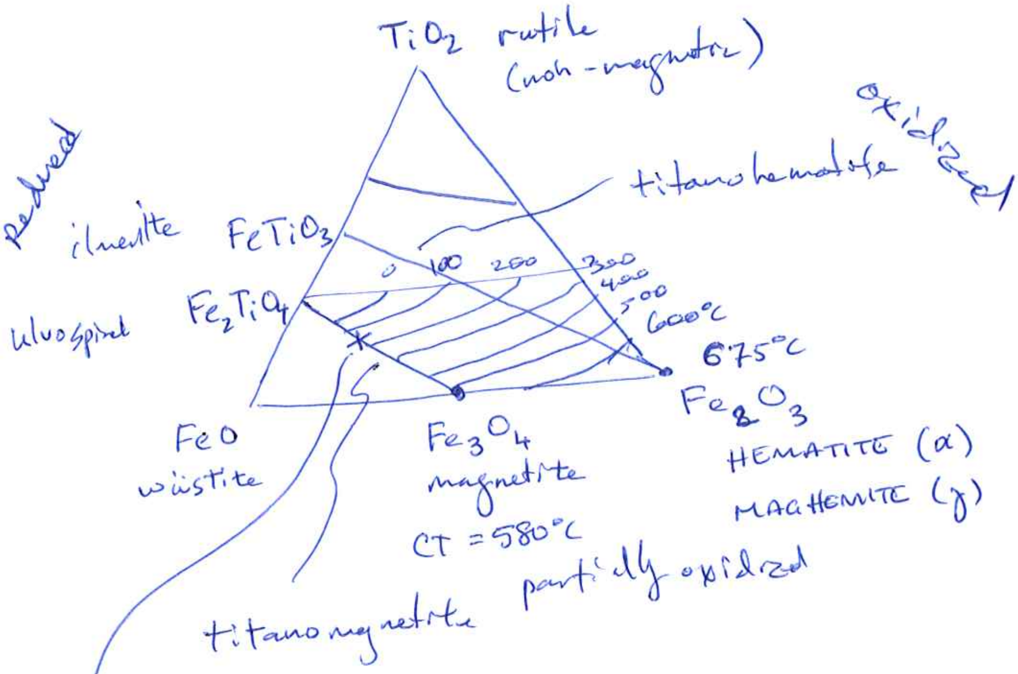
$\mu = \mu_0(1+\chi)$ magnetic permeability

$$\vec{B} = \mu \vec{H}$$

FOR MOST GEOLOGIC MATERIALS $\mu \approx \mu_0$, EXCEPT FOR MAGNETIC MINERALS.

MAGNETIC MINERALS

MOST COMMON, Fe - Ti OXIDES



titanomagnetite $Fe_{2.4}Ti_{0.6}O_4$

most common ~~ocean~~ magnetic mineral in ocean

		avg.
Fe_3O_4 - magnetite	susceptibility	6
Fe_2O_3 - hematite		6.5×10^{-3}

LARGE RANGE in susceptibility highly variable